

SHOPSPHERE CONVERSION INTELLIGENCE

Capstone Pre-Submission Project Report | E-Commerce Analytics Domain
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🔴 Document Scope Note: This pre-submission report reflects the team's preliminary understanding and proposed project direction; refinements and revisions may be incorporated during project execution.

83.8%

Observed conversion rate vs. 1–5% industry-typical — flagged for validation

50,000

Clean sessions retained after removing 500 duplicate records

5

Legacy/ID columns proven non-informative and removed

33

Validated features carried forward into modeling

1. Business Problem Understanding and Objective Formulation

ShopSphere has expanded digital marketing spend across search, social, affiliate, display, and email. Despite traffic growth, leadership faces rising Customer Acquisition Costs (CAC) and declining Return on Ad Spend (ROAS). Aggregated reporting explains macro session counts but fails to explain session-level conversion drivers.

- **Business Objective:** Identify customer, behavioral, and campaign drivers to target high-propensity user segments and optimize CAC/ROAS spend efficiency.
- **Analytical Objective:** Develop an interpretable session-level conversion propensity model (predicting `conversion_flag`), rank drivers via SHAP, and deploy autoencoder-based anomaly detection.

⚠ Critical Discrepancy — Stated Narrative vs. Observed Target Data

The business narrative describes a low-conversion problem, but dataset profiling yields an 83.8% conversion rate (~20–80× higher than typical 1–5% baselines). Conversion flag likely records an upstream engagement event rather than final purchase completion. A mandatory validation audit against raw web analytics is required prior to production rollout.

2. Challenges and Constraints

Beyond the target-definition discrepancy above, the following risks and constraints were identified and are addressed with specific, testable mitigations rather than assumed away:

Identified Risk / Constraint	Impact	Proposed Mitigation Strategy
Target Definition Ambiguity (83.8% rate)	HIGH	Train on <code>conversion_flag</code> while executing a mandatory raw web analytics audit before production deployment.
Data Leakage via <code>cart_additions</code> — shows a near step-function jump in conversion rate (~51.7% at zero vs. ~90% at one or more), consistent with but not confirming leakage	HIGH	Build dual pipelines: Model A (full feature set) and Model B (early, pre-cart features) to test — not assume — whether <code>cart_additions</code> constitutes leakage.
Missing Data (AOV 9.7%, Rating 10%)	MEDIUM	Apply median imputation and generate binary missingness-indicator features.

Absence of Direct Campaign Spend Data	MEDIUM	Deliver relative efficiency rankings and propensity-based ROI guidance rather than absolute monetary estimates.
Business Adoption — stakeholders must trust model-driven targeting over familiar channel-level reporting	MEDIUM	SHAP-based explainability built into the pipeline from the outset, so every recommendation is traceable to a specific, auditable driver.

3. Background Study / Literature Review

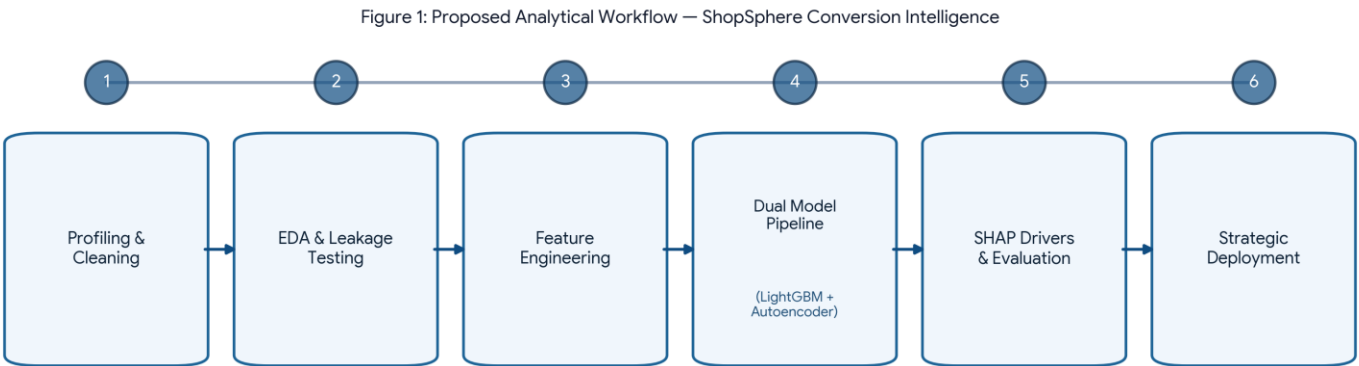
Conversion prediction from clickstream data is well studied in e-commerce analytics. Tagliabue et al. emphasize distinguishing ‘buyers’ from ‘window shoppers’ under clickstream behavioral noise. While Koehn et al. apply RNNs to sequential sessions, 2026 industry benchmarks confirm gradient-boosted decision trees (LightGBM/XGBoost) equal or exceed deep sequence architectures on aggregated session features while offering more accessible SHAP interpretability. Attribution studies further confirm that session-level propensity scoring provides more actionable spend guidance than aggregate channel attribution alone.

4. Proposed Methodology / Solution Approach

The project follows a structured lifecycle from data understanding through business recommendations (Figure 1). Two methodology choices are treated as testable rather than assumed:

- **Model architecture:** Gradient Boosted Trees (LightGBM/XGBoost) combined with SHAP feature attribution are hypothesized to achieve superior discriminative performance over a Logistic Regression baseline without sacrificing driver transparency — both will be trained and compared, not assumed in advance.
- **Anomaly detection:** An autoencoder trained on numerical session metrics is hypothesized to isolate extreme reconstruction-loss outliers corresponding to non-human or corrupted event streams (e.g., bot traffic), providing a complementary check beyond classification alone.

Together, this approach directly addresses the identified business problem: ranked, explainable conversion drivers replace guesswork about which channels or segments to prioritize, giving Marketing and Finance an evidence-based basis for reallocating spend rather than continuing to optimize channel mix with no clear signal of what actually works.



5. Proposed Timeline

Phase	Core Deliverables & Activities	Effort
Week 1	Data profiling, cleaning, duplicate removal, assumptions log	5–6 hrs
Week 1–2	EDA, bivariate analysis, cart-leakage hypothesis testing	6 hrs
Week 2–4	Feature engineering; LightGBM + Autoencoder training; SHAP analysis; packaging	22 hrs

6. Proposed Budget / Resource Requirements

Estimates assume moving this prototype toward a production-ready, scalable deployment, not the coursework prototype itself, which requires no incremental spend beyond a personal laptop and free-tier tools.

Item	Type	Basis
Raw web analytics validation audit	One-time	Required to resolve the Critical Discrepancy (Section 1) before any production trust in conversion_flag
Cloud ML compute (training/retraining)	Recurring	AWS/GCP managed compute for LightGBM + Autoencoder training and periodic retraining
Model hosting / low-latency scoring API	Recurring	Real-time propensity scoring integrated with targeting systems
Data warehousing	Recurring	Session-level data storage at marketplace scale
CRM / ad-platform integration engineering	One-time	Connecting propensity scores to existing targeting/CRM systems
Monitoring & drift detection tooling	Recurring	Given target-definition uncertainty flagged in Section 1

7. Expected Deliverables and Outputs

Core Deliverables Package

Cleaned dataset (50k records), EDA report with leakage tests, trained LightGBM + Autoencoder models, SHAP driver plots, business recommendations report, deployment readiness proposal, executive slide deck, technical report, and GitHub code repository.

Quantitative Success Metrics

- **Primary metric:** Precision-Recall AUC (PR-AUC) and Recall on the non-converted (minority) class — chosen over raw accuracy because the dataset's imbalance favors the converted class (83.8%), making accuracy and even ROC-AUC potentially misleading for the actual business decision (identifying low-propensity sessions worth re-targeting).
- **Secondary metric:** ROC-AUC ≥ 0.85 , reported alongside PR-AUC for comparability with standard classification benchmarks.
- **Unsupervised Anomaly Detection:** Reconstruction Mean Squared Error (MSE) thresholding at the 95th/99th percentile to isolate bot traffic.